Probability Distributions: CDF, PMF, PDF

Math Foundations — Node 23

1 Setup

Throughout, $(\Omega, \mathcal{F}, P)$ is a probability space and $X : \Omega \to \mathbb{R}$ is a real-valued random variable, i.e. a measurable map with respect to the Borel σ-algebra $\mathcal{B}(\mathbb{R})$. The *distribution* (or *law*) of X is the pushforward measure $\mu_X(A) := P(X \in A)$ on $\mathcal{B}(\mathbb{R})$.

2 Cumulative Distribution Function

Definition 1 (CDF)**.** *The* cumulative distribution function *of X is*

$$F_X(x) := P(X \le x) = \mu_X((-\infty, x]), \qquad x \in \mathbb{R}.$$

Theorem 1 (Properties of the CDF)**.** *The CDF F_X of any real-valued random variable X satisfies:*

(i) ***Non-decreasing:*** $x \le y \implies F_X(x) \le F_X(y)$.

(ii) ***Right-continuous:*** $\lim_{y \downarrow x} F_X(y) = F_X(x)$ *for every* $x \in \mathbb{R}$.

(iii) ***Limits:*** $\lim\limits_{x \to -\infty} F_X(x) = 0$ *and* $\lim\limits_{x \to +\infty} F_X(x) = 1$.

Proof. (i) Monotonicity. If $x \le y$, then $\{X \le x\} \subseteq \{X \le y\}$, so by monotonicity of the probability measure,

$$F_X(x) = P(X \le x) \le P(X \le y) = F_X(y).$$

(ii) Right-continuity. Fix $x \in \mathbb{R}$ and let $(y_n)_{n \ge 1}$ be any sequence with $y_n \downarrow x$. Define $A_n := \{X \le y_n\}$. Since y_n is decreasing, $A_1 \supseteq A_2 \supseteq \cdots$, and

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} \{X \le y_n\} = \{X \le x\},$$

because $y_n \to x$ from above implies $X(\omega) \le y_n$ for all n iff $X(\omega) \le \inf_n y_n = x$. The sets A_n have finite measure (bounded by 1), so continuity from above of P (a consequence of countable additivity) gives

$$\lim_{n \to \infty} P(A_n) = P\Big(\bigcap_{n=1}^{\infty} A_n\Big) = P(X \le x) = F_X(x).$$

Hence $\lim_{n \to \infty} F_X(y_n) = F_X(x)$. Since this holds for every such sequence, F_X is right-continuous at x.

(iii) Limits at $\pm\infty$. Take any sequence $x_n \uparrow +\infty$ and let $B_n := \{X \le x_n\}$. Then $B_1 \subseteq B_2 \subseteq \cdots$ and

$$\bigcup_{n=1}^{\infty} B_n = \{X < +\infty\} = \Omega,$$

since X is real-valued. By continuity from below of P (again, countable additivity),

$$\lim_{n\to\infty} F_X(x_n) = \lim_{n\to\infty} P(B_n) = P(\Omega) = 1.$$

Similarly, take $x_n \downarrow -\infty$ and $C_n := \{X \le x_n\}$. Then $C_1 \supseteq C_2 \supseteq \cdots$ and

$$\bigcap_{n=1}^{\infty} C_n = \{X = -\infty\} = \emptyset,$$

so by continuity from above,

$$\lim_{n\to\infty} F_X(x_n) = P(\emptyset) = 0.$$

This proves all three properties. □

Remark 1. *The converse also holds: every function $F : \mathbb{R} \to [0,1]$ satisfying (i)–(iii) is the CDF of some random variable. (Construction: take $\Omega = (0,1)$ with Lebesgue measure and define $X(u) = \inf\{x : F(x) \ge u\}$, the generalized inverse. This is the inverse-CDF sampling method.)*

3 Probability Mass Function (Discrete Case)

Definition 2 (PMF)**.** *A random variable X is* discrete *if there exists a countable set $S \subset \mathbb{R}$ with $P(X \in S) = 1$. Its* probability mass function *is*

$$p_X(k) := P(X = k), \qquad k \in S.$$

It satisfies $p_X(k) \ge 0$ and $\sum_{k\in S} p_X(k) = 1$, and the CDF is the running sum

$$F_X(x) = \sum_{k\in S,\, k\le x} p_X(k).$$

The CDF of a discrete RV is a step function with jumps of size $p_X(k)$ at each $k \in S$.

4 Probability Density Function (Continuous Case)

Definition 3 (PDF)**.** *A random variable X is* absolutely continuous *if there exists a measurable function $f_X : \mathbb{R} \to [0,\infty)$ such that*

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt \quad \textit{for all } x \in \mathbb{R}.$$

Such an f_X is called a probability density function *of X and satisfies $\int_{\mathbb{R}} f_X = 1$. It is unique up to a Lebesgue-null set, and where F_X is differentiable, $f_X(x) = F_X'(x)$.*

For absolutely continuous X and any $a < b$,

$$P(a < X \le b) = F_X(b) - F_X(a) = \int_a^b f_X(t)\,dt.$$

In particular $P(X = x) = 0$ for every single point x, so $f_X(x)$ is a *density*, not a probability — it can exceed 1 (e.g. Uniform$(0, 0.5)$ has $f_X = 2$ on $[0, 0.5]$).